Initial acquisition and sharper count comparison for fixed-target Berry–Esseen excess

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Abstract

We continue the fixed-target approach to the sharp iid Berry–Esseen problem. The first support and moment bounds use fixed-target contact, an external structural inequality and the sharp predecessor bound. No upper bound on the actual score is needed. A moment-sensitive Gaussian-weight condition then permits a small complete scalar cover of the initial moment interval. Retaining the phase cost improves its defect bound, although the remaining bound is too large for the next general contraction step. Separately, disjoint crossing events and a fourth-moment estimate reduce the cost in the conditional-count comparison. A combined moment response gives a further conditional improvement on one central domain. We give the new proofs and distinguish them from the retained Fourier estimates, domain-acquisition premises and finite-base obligations. No complete sharp upper bound is claimed.

How the pieces fit. The earlier companion establishes attainment, contact and conservation for the excess $\sqrt{n}(F_n - \Phi) - C_{\mathrm{E}}\rho$. Here we first make its initial entry quantitative, then calculate what that entry actually yields. The count-crossing argument and the common moment response are separate local improvements; neither can be used before its stated domain is acquired. The proof of each implication is given below. The low- and high-frequency inequalities named in Section 4 remain explicit analytic inputs, as in the full manuscript. Exact evaluation of their formulas does not independently prove Fourier inversion or their uniform remainder bounds.

The external affine inequality used for initial entry is Shevtsova's 2013 structural bound, valid for every sample size; see *Informatics and its Applications* **7**(1), 124–125, `https://www.mathnet.ru/eng/ia252`. It is applied after variance normalization, where the third-moment sum is $\rho/\sqrt{n}$ and the sum of cubed standard deviations is $1/\sqrt{n}$. Its uniform bound also applies to closed CDF values by taking one-sided limits.

1 Entry from fixed-target excess

This note isolates the first acquisition step. It does not assert a complete local exclusion or a sharp Berry–Esseen theorem. Let P be a centered, unit-variance law, S_j a sum of j independent copies, and set

$$\rho = \mathbb{E}|X|^3, \quad \beta = \mathbb{E}[X|X|], \quad \gamma = \mathbb{E}X^3, \quad c = C_{\mathrm{E}} = \frac{3+\sqrt{10}}{6\sqrt{2\pi}}.$$

The threshold H always has the units of the raw sum. A fixed-target maximizer maximizes

$$a(P,H) = \sqrt{n}\{\mathbb{P}(S_n \le H) - \Phi(H/\sqrt{n})\} - c\rho.$$

We use the attainment and contact theorem in the companion fixed-target note. In particular its closed-support contact equality, not ratio optimality, is the variational premise here. We consider a maximizer with $a \ge 0$. The actual normalized score is $c + a/\rho$; no upper bound on it is assumed.

Theorem 1.1 (Initial moment and support entry)**.** *Assume $n \geq 42$, $C_{n-1} \leq c$, and the universal structural inequality*

$$\sqrt{n}\,\Delta_n(P) \leq \frac{3031}{10000}\left(\rho + \frac{323}{500}\right) \tag{1}$$

for standardized laws with finite third moment. Then a fixed-target maximizer with $a \geq 0$ satisfies

$$1 \leq \rho < \frac{7361}{4000} < \frac{1841}{1000}, \qquad |\beta| < \frac{321}{500}, \qquad \operatorname{supp} P \subset (-96/25, 96/25). \tag{2}$$

The theorem is uniform in all real H and in both signs of a support point. It requires neither C_{n-2} nor an excess bound of order $1/n$.

Inequality (1) is an external structural input, recorded with its Shevtsova attribution in the manuscript introduction. It is not proved by the exact arithmetic below. Only this one of the two affine structural bounds is needed for the displayed moment radius.

Proof. Since $c\rho \leq \sqrt{n}\Delta_n(P)$ and $c > 819/2000$, elementary division gives

$$\rho < \frac{(3031/10000)(323/500)}{819/2000 - 3031/10000} = \frac{7361}{4000}.$$

The other affine bound $0.3322(\rho + 0.429)$ alone would give the larger number $712569/386500$; it must not be used to claim $7361/4000$. Put $m = \mathbb{E}X_+ = \mathbb{E}X_- > 0$ and $a_\pm = \mathbb{E}X_\pm^2 = (1 \pm \beta)/2 > 0$. Cauchy on each sign class gives $m\rho \geq a_+^2 + a_-^2$ and $m^2 \leq a_+ a_-$. Hence

$$\rho \geq \frac{1 + \beta^2}{\sqrt{1 - \beta^2}}.$$

This function increases with $|\beta|$. For $r = 1841/1000$, $b = 321/500$,

$$(1 + b^2)^2 - r^2(1 - b^2) = \frac{93288849}{50000000000} > 0,$$

which proves the second assertion without assuming bounded support.

Write $v = n-1$, $s = \sqrt{n/v}$, $F_v(x) = \Phi(x/\sqrt{v})$ and $e_n = n(s-1) - 1/2$. Closed-support contact, the upper predecessor bound $F_{n-1,P}(H-t) \leq F_v(H-t) + c\rho/\sqrt{v}$, and $\sqrt{n}(F_{n,P}(H) - \Phi(H/\sqrt{n})) = c\rho + a$ give

$$0 \leq H_n(H,t) + c\{-|t|^3 + \tfrac{3}{2}\rho t^2 + 3\beta t + \rho e_n\} - na \quad (t \in \operatorname{supp} P), \tag{3}$$

where exactly

$$\frac{H_n(H,t)}{n^{3/2}} = F_v(H-t) - F_{v+1}(H) + tF'_{v+1}(H) - \frac{t^2 - 1}{2}F''_{v+1}(H).$$

The extra $-na$ can be discarded in this upper bound. This is the precise reason the predecessor assumption is $C_{n-1} \leq c$, not a comparison with the actual score. No continuity assumption on the distribution function at $H - t$ is inserted.

Rationalization gives

$$ne_n = \frac{s^2(2s+1)}{2(s+1)^2}, \qquad s < 81/80 =: S, \qquad ne_n < \frac{793881}{2073680} < \frac{48}{125}.$$

The first expression increases for $s \geq 1$, as its logarithmic derivative is $2/s + 2/(2s+1) - 2/(s+1) > 0$. Thus $e_n < 8/875$.

For completeness we rederive the Gaussian estimate used at the radius. The heat identity $\partial_v F_v = F_v''/2$ permits spatial expansion of $F_v(H-t)$ to order four, variance expansion of F_{v+1} to

order two, and variance expansion of F'_{v+1}, F''_{v+1} to order one. The cubic coefficient is $(-t^3+3t)/6$, and the quartic coefficient is $(t^4-6t^2+3)/24$. Therefore, for $t=\pm u$, $z_v = H/\sqrt{v}$,

$$H_n(H,\pm u)/s^3 = \pm A(1-z_v^2)\phi(z_v) + B\phi'''(z_v) + \mathcal{E}, \tag{4}$$

where $A=(u^3-3u)/6$, $H_4 = u^4-6u^2+3$ and $B = H_4/(24\sqrt{v})$. The four error bounds after division by s^3 are respectively

$$\frac{u^5\|\phi^{(4)}\|_\infty}{120v}, \quad \frac{\|\phi^{(5)}\|_\infty}{48v^{3/2}}, \quad \frac{u\|\phi^{(4)}\|_\infty}{8v}, \quad \frac{(u^2-1)\|\phi^{(5)}\|_\infty}{16v^{3/2}}.$$

Gaussian Fourier inversion gives norms at most $3\phi(0) < 6/5$ and $8/\pi < 51/20$. Since $v \geq 41$ and $\sqrt{v} > 6$, for $u \geq 5/2$ this gives

$$|\mathcal{E}| < E(u) := u^5/4100 + 3u/820 + 17(3u^2-2)/78720.$$

These errors are uniform in H; they use no higher moment of P.

If $A, B > 0$ and $B \leq A/4$, the leading part of (4) is at most $\phi(0)(A + 9B^2/(4A))$ for both signs. Indeed, for the positive sign and $|z_v| \leq 1$, it is at most $\phi(0)(A - Az_v^2 + 3B|z_v|)$, and complete the square. Outside that interval the first term is nonpositive. For the negative sign the first term is at most $2e^{-3/2}A\phi(0) < A\phi(0)/2$. In both cases use $\|\phi'''\|_\infty < 2\phi(0)$: on $[0,\sqrt{3}]$ the polynomial $3x-x^3$ is at most two, and beyond $\sqrt{3}$ use $x^3e^{-x^2/2} \leq 3\sqrt{3}e^{-3/2} < 2$.

For frozen r, b above define

$$Q(u) = -u^3 + \tfrac{3}{2}ru^2 + 3bu + (8/875)r,$$
$$P(u) = \tfrac{2}{5}S^3\left(A^2 + \frac{H_4^2}{256\cdot 41}\right) + S^3E(u)A + \frac{819}{2000}Q(u)A.$$

At $R = 96/25 < 6$ we have $A > 0$, $0 < H_4 < 36A$, hence $B < A/4$. The sign $Q(R) < 0$ is checked before replacing c by its lower endpoint. The upper bound in (3), multiplied by A, is then strictly below $P(R)$, and exact arithmetic gives

$$-P(R) = \frac{2214258283536204102723}{10250000000000000000000} > \frac{1}{5}.$$

To cover each entire exterior ray, differentiation of the exact Gaussian expression gives

$$|\partial_t H_n(H,t)| < \tfrac{2}{5}(t^2/2 + 1/2 + e_n).$$

For verification put $y = (H-t)/\sqrt{n}$, $z = H/\sqrt{n}$; the derivative is $n\phi(z) + Ht\phi(z) - ns\phi(sy)$. The second-order Taylor error between $\phi(y)$ and $\phi(z)$ costs $\phi(0)t^2/2$, while $\partial_s[s\phi(sy)] = -\phi''(sy)$ costs $\phi(0)n(s-1)$. Define $L(u) = \frac{2}{5}(u^2/2 + 1/2 + 8/875) + (819/2000)Q'(u)$. At R, the six numbers Q, Q', Q'', L, L', L'' are strictly negative. Since $Q''' = -6$ and $L'' = 2/5 - 6(819/2000) < 0$, all their required signs persist for $u \geq R$. Thus both radial upper bounds decrease from a strictly negative boundary value. The contact inequality excludes every support point with $|t| \geq R$. □

2 Moments and an actual high-frequency event

Higher moments are used only after Theorem 1.1. For $x = |X|$ the nonnegative measure $d\mu_0 = (x-1)^2(x+1/2)\,dP$ has mass $\rho - 1$. Direct expansion gives

$$M_4 - 1 = \int \frac{(x+1)^2}{x+1/2}\,d\mu_0,$$
$$M_5 - 1 = \int \frac{x^3+2x^2+3x+3/2}{x+1/2}\,d\mu_0,$$

where $M_j = \mathbb{E}|X|^j$. Both ratios increase for $x \geq 0$: they equal $x + 3/2 + 1/[4(x + 1/2)]$ and $x^2 + 3x/2 + 9/4 + 3/[8(x + 1/2)]$, whose derivatives are nonnegative. Writing their values at $R = 96/25$ as $k_4(R), k_5(R)$, we obtain

$$M_4 \leq \min(R\rho, 1 + k_4(R)(\rho - 1)), \quad M_5 \leq \min(RM_4, 1 + k_5(R)(\rho - 1)), \quad M_6 \leq RM_5. \tag{5}$$

Also $\gamma^2 \leq M_4 - 1$, by expanding $\mathbb{E}(X^2 - 1 - \gamma X)^2 \geq 0$.

There is no need to assume a central support gap before the first Fourier bound. For independent copies let $E = \{(X - X')^2 \geq 9/25\}$ and let M be any valid fourth-moment ceiling. Then

$$\mathbb{P}(E) \geq \alpha := \frac{(2 - 9/25)^2}{2M + 6} > 0, \quad |X - X'| \geq 3/5 \text{ on } E, \quad \mathbb{E}[(X - X')^2 \mid E] \leq 2/\alpha. \tag{6}$$

Indeed $\mathbb{E}[(X - X')^2; E] \geq 2 - 9/25$ and $\mathbb{E}(X - X')^4 = 2M_4 + 6 \leq 2M + 6$, so Cauchy proves the mass bound. The conditional second-moment bound follows from its unconditional value two. This is an event under one actual iid marginal, not a replacement characteristic function.

3 A sharper first license

The retained fixed-target conserved identity is

$$(n + H^2)a + nc\rho - \frac{n\phi(H/\sqrt{n})}{6}(\gamma + 3J) + n^{3/2}\mathcal{L} = \mathcal{R}_c(H). \tag{7}$$

Put $\tau(t) = \mathbb{E}[X\mathbf{1}_{X \geq t}]$ and $\nu = \mathcal{L}(H - S_{n-1})$. Over complementary support gaps, define

$$J = \sum \tau_{u,v}(v - u)^2, \qquad \mathcal{L} = \sum \tau_{u,v} \int_{[u,v)} (t - u)\, d\nu(t) \geq 0.$$

The half-open interval includes the correct endpoint atoms. The elementary gap inequality gives $D = (3 + \sqrt{10})\rho - \gamma - 3J \geq 0$. In particular the right side of (7) is nonnegative when $a \geq 0$, since it can also be written as

$$(n + H^2)a + nc\rho(1 - e^{-w}) + n\phi(0)e^{-w}D/6 + n^{3/2}\mathcal{L}, \qquad w = H^2/(2n).$$

Lemma 3.1 (Moment-cell license)**.** *Under* (7)*, with* $D, \mathcal{L} \geq 0$*, suppose* $n \geq N$*,* $\rho \in [l, u]$*,* $a \geq 0$*, and* $0 < c_- \leq c$*. If a bound valid for every real* H *gives*

$$\mathcal{R}_c(H) \leq T < Nc_-l, \tag{8}$$

then either $T < 0$ *and this cell is excluded, or*

$$\gamma + 3J > 0, \qquad e^{-H^2/(2n)} \geq 1 - \frac{T}{Nc_-l} > 0, \tag{9}$$

$$(n + H^2)a + \frac{n\phi(0)}{6}D + n^{3/2}\mathcal{L} \leq \mathcal{R}_c(H) \leq T. \tag{10}$$

Thus the first gate need not be the stronger uniform condition $T < 2N/5$*.*

Proof. For a nonexcluded cell $T \geq 0$. Divide (7) by n:

$$\frac{\phi(H/\sqrt{n})}{6}(\gamma + 3J) = (1 + H^2/n)a + c\rho + \sqrt{n}\mathcal{L} - \mathcal{R}_c/n \geq c\rho - T/n > 0.$$

This proves the sign. Since $D \geq 0$, $\gamma + 3J \leq 6c\rho/\phi(0)$; hence $c\rho e^{-w} \geq c\rho - T/n$, proving (9). Finally rewrite the conserved expression as the left side of (10), plus $n(1 - e^{-w})(c\rho - \phi(0)D/6)$. Its last coefficient is $\phi(0)(\gamma + 3J)/6 > 0$, so it can be dropped. □

The positivity argument uses no small-defect premise. The input $D \geq 0$ is used for the density and unweighted-resource conclusions, not for the first sign deduction. Thresholds of either sign and $H = 0$ are included. If only $T \leq Nc_l$ is proved, the strict sign and positive density floor do not follow. The strict gate is a genuine acceptance condition.

Lemma 3.2 (First phase price without a small defect)**.** *Under Lemma 3.1, put*

$$\mathcal{U} = (n + H^2)a + n\phi(0)D/6 + n^{3/2}\mathcal{L}, \qquad q_{\mathrm{ph}} = \tfrac{1}{2}(c_l - T/N) > 0.$$

Then $\mathcal{U} \leq \mathcal{R}_c(H) - q_{\mathrm{ph}}H^2$*. In particular a complete polynomial bound* $\mathcal{R}_c(H) \leq A + F|H| - qH^2$ *with* $F \geq 0, q > 0$ *gives*

$$\mathcal{U} \leq A + \frac{F^2}{4(q + q_{\mathrm{ph}})}.$$

This improved bound is on $\mathcal{U}$*, not on the raw* $\mathcal{R}_c$*.*

Proof. Let $M_c = c\rho - \phi(0)D/6$ and $w = H^2/(2n)$. The exact identity and $\mathcal{R}_c \leq T$ imply $e^{-w}M_c \geq c\rho - T/n > 0$. Therefore

$$n(1 - e^{-w})M_c = n(e^w - 1)e^{-w}M_c \geq \frac{H^2}{2}(c\rho - T/n) \geq q_{\mathrm{ph}}H^2.$$

Here $e^w - 1 \geq w$ for $w \geq 0$. Subtract this term from the exact decomposition $\mathcal{R}_c = \mathcal{U} + n(1 - e^{-w})M_c$, then complete the square. There is no premise $D < 2$, and no multiplication of separate lower estimates on M_c and e^{-w}. □

Lemma 3.3 (Normalized first-cover consequence)**.** *Under the preceding hypotheses, suppose a separate complete enclosure proves* $\mathcal{R}_c(H)/\rho < 10$ *for all real* H*. For* $n \geq 42$*,*

$$e^{-H^2/(2n)} > \frac{7199}{17199} > \frac{209}{500}, \qquad \frac{D}{\rho} < \frac{25000}{6979}, \qquad \frac{a}{\rho} < \frac{10}{n}.$$

In particular the first license is discharged. The actual score satisfies $c + a/\rho < c + 10/n$*, but this is an output, not an entry premise.*

Proof. Use the pointwise version of the preceding proof with $T = 10\rho$: $10 < 42(819/2000) < nc$. The density floor is strictly larger than $1 - 10/(42c) > 1 - 10/[42(819/2000)] = 7199/17199$. From (10) and $\phi(0) > 997/2500$, $D/\rho < 60/(42\phi(0)) < 25000/6979$. The nonnegative term $(n + H^2)a$ gives the final bound. □

4 The first scalar enclosure and its remaining gate

This section states exactly what a bounded arithmetic cover must supply. It proves the algebraic reduction, rather than treating an archived PASS as an analytic premise. Fix a closed cell $[l, u] \subset [1, 7361/4000]$. Choose outward ceilings M, g, E_5, b for $M_4, |\gamma|, M_5, |\beta|$ using (5), $|\gamma| \leq \rho$, Pearson, and the sign-moment inequality. In particular $M < 11.1$, $g < 1.85$ and $M_6 \leq RE_5$.

Write $J_1 = \sum \tau_{a,b} \int_a^b (t - a)t\, dt$ and $J_2 = \sum \tau_{a,b} \int_a^b (t - a)t|t|\, dt$. Set

$$K_1 = \mathbb{E}[X|X|^3] - 3\beta - 3\rho\gamma/2,$$
$$K_q = M_5/4 - \rho M_4/2 - 3\beta\gamma/2 + \rho/2, \qquad U = J_1 + (M_4 - 3)/6.$$

The exact raw remainder in (7) is

$$\mathcal{R}_c = c(3\beta J/2 + 3\rho J_1 - 3J_2 - K_q) + H(cK_1 + \phi U) - H^2(c\rho + \gamma\phi/2) + E_n.$$

The fixed target c is the coefficient in every term. Its enclosing interval $[819/2000, 20487/50000]$ is not a bound on the actual score.

The elementary gap integrals give $-M_4/6 \le J_1 \le M_4/2$, $-3J_2 - M_5/4 \le 0$ and $J \le \rho + 1$. The first two follow by integrating, gap by gap,

$$\int_a^b (t-a)t\,dt + \tfrac{1}{2}\int_a^b t^2 dt = (b-a)b^2/2,$$

and its complementary upper inequality; for the negative part of J_2 use $\int_u^v (v-x)x^2dx \le \frac{1}{3}\int_u^v x^3dx$. Consequently a valid constant bound is

$$A_2 = \tfrac{3}{2}b(2u+1) + 2uM - l/2.$$

One may take the smaller of A_2 and

$$A_1 = \tfrac{3}{2}b(2u+1) + \tfrac{3}{2}\Gamma(R,l)(2u) + \tfrac{3}{8}u^2(u+1) + \tfrac{8}{27}u^3 - l/2,$$

where $\Gamma(R,l)$ bounds the nonnegative part of

$$-v^2/4 + (R+l)v/3 - Rl/2 + l^3(2R-l)/(12v^2), \qquad l \le v \le R.$$

This is $v^{-2}\int_l^v (R-x)x(x-l)\,dx$. For each realized negative magnitude v, Tonelli bounds all its gap contributions by this one integral; multiplying by v and averaging charges $\Gamma\mathbb{E}X_-^3$. Thus on negative gaps it charges $3(\rho J_1 - J_2)$ by $3\Gamma(\rho-\gamma)/2$; on positive parts use $t(\rho-t) \le \rho^2/4$, multiply by $(t-a) \ge 0$, and integrate $\sum \tau_{a,b}\int_a^b (t-a)\,dt = J/2$. This gives $3\rho^2 J/8$ and includes the positive part of a central gap with $a < 0$. The remaining term uses $-K_q \le 8\rho^3/27 + (3b/2 - 1/2)\rho$, obtained by $M_5 \ge M_4^{3/2}$ and maximizing $\rho y^2/2 - y^3/4$. Thus $A_* := \min(A_1, A_2)$ is valid when the chosen Γ is certified.

Valid bounds on $|K_1|$ are

$$\begin{aligned} k_{\mathrm{rad}} &= 3ub + \frac{R(R+2-3l/2)}{R+1/2}(u-1), \\ k_{\mathrm{old}} &= 6u - 3u^2/2 + 3b, \\ k_R &= Ru/2 + \max_{t=l,u} |R/2 - 3t/2|g + 3b. \end{aligned}$$

For the first, use the radial identity for $K_1 + 3\rho\beta$ with ratio $x(x+2-3\rho/2)/(x+1/2)$, convex in x and decreasing in ρ. Its negative part is at most $31/40$, so require its stated endpoint ceiling to be at least $31/40$. The second uses $R \le 6$ and $|\mathbb{E}[X|X|^3] - 3\gamma| \le 3\rho$. The third bounds the two fourth semimoments by $R(\rho \pm \gamma)/2$. Take k to be their minimum. Also $U \in [-1/2, 2M/3 - 1/2]$, so use $v_H = (2/5)\max(1/2, 2M/3 - 1/2)$ (or a separately proved smaller bound). Then $q_* := c_- l - g/5 > 0$ is a valid quadratic curvature.

It remains to have a *complete* Fourier bound

$$|E_n| \le K_F + s_F|H|, \quad K_F = \tfrac{16}{25}(L_0 + K_H + L_2/\sqrt{N}), \quad s_F = \tfrac{16}{25}L_H. \tag{11}$$

The compensated quadratic-hinge inversion has factor $2\sqrt{n}/\pi$. The retained triangle transfer uses the parameters N, M, g, M_6 and sixth-order Taylor estimates on $[0, 1/2]$. Its conclusion is

$$I_0 \le (L_0 + L_H|H|)/\sqrt{n} + L_2/n.$$

The retained complete six-part high-frequency lemma uses the actual event (6) and gives $\sqrt{n}I_\infty \le K_H$. Every damping, floor and window gate in that lemma must hold; one takes a minimum only of complete six-term bounds. The event proof does not permit dropping its Gaussian comparator tail. Neither Fourier lemma uses a score bound or a defect feedback hypothesis.

Completion of the square now proves the all-threshold bound

$$\mathcal{R}_c(H) \le T := K_F + c_+ A_* + \frac{(c_+ k + v_H + s_F)^2}{4q_*}, \tag{12}$$

provided $A_* \ge 0$, $k \ge 0$, $q_* > 0$ and $c \in [c_-, c_+]$. This is a scalar inequality, so using a formal coefficient interval containing c is sound independently of the actual score. The first numerical obligation is a finite closed cover of $[1, 7361/4000]$ on which the stated Fourier gates and $T < Nc_l$ all hold. The endpoint $\rho = 1$ is included; at that point the law is a symmetric two-point law, and all moment ceilings remain well-defined. Adjacent cell endpoints must both be covered.

Scope of this note. The bounded companion checker verifies the support rational reductions, five exact finite-law moment tests, and a negative control rejecting radius three with the same envelope. It does not evaluate a whole raw cover or prove any local comparison predicate. A separate 24-cell calculation reports $\mathcal{R}_c/\rho < 10$; its source, outward arithmetic, coverage and complete Fourier premises require their own review before composition with Lemma 3.3. Subsequent support contractions, central-gap acquisition, confinement and LOW/HIGH exclusion still require their own ordered hypotheses. In particular the present entry theorem alone does not remove the independent finite base.

5 What the first cover proves

The numerical obligation in Section 4 is small enough to test directly. Take $N = 42$, $R = 96/25$ and the 24 closed intervals with endpoints

$$r_i = 1 + \frac{3361i}{96000}, \qquad 0 \le i \le 24.$$

Adjacent intervals share their endpoint; no boundary or sign class is discarded. Enclose the fixed coefficient, using Machin's formula and rational square-root bounds, by

$$c_- = 0.4097321837023 < C_{\mathrm{E}} < c_+ = 0.4097321837025.$$

These numbers do not enclose the maximizing law's actual score.

On $[l, u]$, use the moment bounds of (5). The low triple is the sixth-order triangle estimate; the high estimate is the minimum of six complete six-term bounds, at windows $1/8, 4/25, 9/50, 1/5, 11/50, 1/4$. Every window's gates are checked before its complete bound is admitted. With the notation of the preceding section, set

$$q(t) = tl - g/5, \quad f(t) = tk + v_H + s_F, \quad B(t) = K_F + tA_* + \frac{f(t)^2}{4q(t)}.$$

For fixed H, the pre-completion expression $K_F + tA_* + f(t)|H| - q(t)H^2$ is affine in t. Its supremum over H is convex. Hence the sharper endpoint choice

$$T_i = \max\{B(c_-), B(c_+)\}$$

is valid for all intermediate t, including C_{E}. All divisors are strictly positive. The retained source functions and the explicit rational reconstruction are listed in `cell.md`; `cell_check.py` computes the full cover afresh.

The exact fractions imply the outward rounded bounds

$$\max_i T_i/r_i < 9.455732088645 < 10, \qquad \min_i (42c_- r_i - T_i) > 13.996057920754. \tag{13}$$

Thus, conditional on the full stated Fourier estimates, the first license holds on the entire initial moment domain. The last cell has $T_{23} < 17.069861855816$, above the old sufficient uniform cutoff $2N/5 = 16.8$. It nevertheless passes the moment-sensitive cutoff. This example explains why the change of gate matters.

5.1 Three ordered refinements

Keep a raw ceiling T, a conserved-resource ceiling U and a defect ceiling d as separate quantities. Initially $U = T$ and $d = 6U/(N\phi_-)$, with $\phi_- = 997/2500 < \phi(0)$. All retained ceilings in the following computation are rounded upward to multiples of 10^{-10}.

At each step, the previous raw ceiling gives

$$\phi_{\min} = \phi_- \left(1 - \frac{T}{Nc_-l}\right) > 0.$$

The gap bound $J \le \rho + 2m$, where $m = \mathbb{E}X_+ \le 1/2$, implies

$$D \ge \sqrt{10}\rho - \gamma - 6m.$$

It is an inequality, not an identity. Thus, with $\sigma_- = 3.162277660 < \sqrt{10}$, the previous defect ceiling yields

$$\gamma_{\min} = \max(-g, \sigma_- l - 3 - d), \qquad \rho - \gamma \le \min(2u, (1 - \sigma_-)l + 3 + d).$$

Use the latter to improve only the negative-gap part of A_1 in Section 4; retain its minimum with the old constant bound. The sign-correct new curvature is

$$q(t) = \max\left(tl - g/5,\ tl + \tfrac{1}{2}\min(\phi_{\min}\gamma_{\min}, (2/5)\gamma_{\min})\right).$$

All bounds are frozen during the endpoint maximum. Both alternatives have slope l in t, so the same convexity argument applies. Recompute the raw upper envelope and take its minimum with the previous T. This new raw bound now licenses the phase price

$$\eta = \tfrac{1}{2}(c_-l - T/N) > 0.$$

Lemma 3.2 bounds the conserved resource by the endpoint envelope with denominator $4(q(t)+\eta)$. Take its minimum with the previous U, then update d by $6U/(N\phi_-)$. Every update uses already justified bounds; the desired defect is never an input to its own proof. In particular, U is never substituted for T in the Gaussian-weight license.

The following are outward upper bounds from exactly three steps, with all 24 cells retained:

Stage	maximum T	maximum U	maximum d	cells with $d < 4$
Initial	17.0698618559	17.0698618559	6.1147234045	—
Step 1	17.0698618559	13.9304183554	4.9901197720	19
Step 2	17.0698618559	13.9304183554	4.9901197720	19
Step 3	17.0698618559	13.9304183554	4.9901197720	19

The last resource bound also gives the weighted statement $nD \le 209.585030422267$; it is not inferred merely from an unweighted bound on D. Some lower moment cells improve at steps 2 and 3, but the worst cell does not. In the final cell the acquired lower bound on γ is still -1.84025.

The five cells numbered 19–23 remain outside the sufficient $D < 4$ regime. A finer bound coupling the moments, or a new structural restriction, is needed there. The first license has been checked; the full initial acquisition and the local exclusion have not. This 24-cell cover must not be compared with the size of the older dense acquisition as though both computations established the same conclusion.

6 A smaller cost for count crossings

The whole-count comparison contains a discrete Taylor remainder of the binomial CDF. Its present bound pays separately for positive and negative crossings, although those events are disjoint. It also uses a third-moment estimate for crossings that can occur only beyond a fixed positive distance. Both losses can be removed without changing the law or the comparison target.

Here is the coupling, including its endpoint convention. Let $U_{0,i}$ and $U_{1,i}$, $1 \le i \le n$, be independent centered residuals, with $U_{0,i} \le a_0$, $U_{1,i} \ge -a_1$ and $c = 1 - a_0 - a_1 \in (1/2, 1]$. Put

$$V_j = \sum_{i \le j} U_{1,i} + \sum_{i > j} U_{0,i}, \qquad 0 \le j \le n,$$

and extend V_j constantly outside $[0, n]$. The sequence $j + V_j$ is strictly increasing, with increments at least c. For a legal count k and $|u| \le 1/2$, define

$$K = \max\{j \in \mathbb{Z} : j + V_j \le k + u\}, \quad R = K - k, \quad t = \mathbb{E}V_k^2.$$

If an independent selector has law $\mathrm{Bin}(n, p)$, its closed event occurs exactly when the selector is at most K. Thus the mixture CDF is $\mathbb{E}F_B(k + R)$. This construction uses the actual conditional laws; it does not replace the convolution by independent choices of moment bounds.

Write $v = p(1-p)$, $\pi_j = \binom{n}{j} p^j (1-p)^{n-j}$, with zero extension, $m_B = \max_j \pi_j$, and $\chi = (n+1)p - k$. Assume $\chi < 2$. For integer r, the binomial difference identity and double summation give

$$F_B(k + r) - F_B(k) - r\pi_k \le \frac{m_B}{(n+1)v} \left\{ (\chi - 1)_+ r_+ + \chi_+ \binom{r_-}{2} + \binom{r_-}{3} \right\}. \tag{14}$$

For completeness, $\pi_j - \pi_{j-1} = \pi_j((n+1)p - j)/(p(n - j + 1))$. When its numerator is positive, the denominator is at least $(n+1)v$. For $r > 0$, the positive differences have weights $r - i + 1$ at $j = k + i$; only $i = 1$ can contribute positively because $\chi < 2$. For $r = -h$, the weights are $h - 1 - i$ at $j = k - i$, $0 \le i \le h - 2$. The bound $(\chi + i)_+ \le \chi_+ + i$ gives the two binomial coefficients in (14). Zero extension handles counts outside $[0, n]$.

Lemma 6.1 (Disjoint costs and a fourth-moment bound)**.** *Put $d = 2c - 1/2 > 0$ and*

$$M_2(\chi, c, u) = \max\left\{ \frac{(\chi - 1)_+}{(c - u)^2}, \frac{\chi_+}{[c + \min(u, c/3)]^2} \right\}.$$

Then, pointwise,

$$(\chi - 1)_+ R_+ + \chi_+ \binom{R_-}{2} \le M_2(\chi, c, u) V_k^2, \tag{15}$$

$$\binom{R_-}{3} \le (V_k)_+^3 / d^3, \qquad \binom{R_-}{3} \le (V_k)_+^4 / d^4. \tag{16}$$

If also $|U_{s,i}| \le A$ and $t \le T$, then

$$\mathbb{E}\{F_B(k + R) - F_B(k) - R\pi_k\} \le \frac{m_B t}{(n+1)v} \left[M_2(\chi, c, u) + \min\left\{ \frac{\sqrt{A^2 + 3T}}{d^3}, \frac{A^2 + 3T}{d^4} \right\} \right]. \tag{17}$$

Proof. On $R = r \ge 1$, crossing monotonicity gives $-V_k \ge cr - u$. The difference $(cr - u)^2 - r(c - u)^2 = (r - 1)(c^2 r - u^2)$ is nonnegative. On $R = -h \le -2$, the count immediately above K is outside the closed event, so $V_k > c(h - 1) + u$. Set $j = h - 1 \ge 1$. For $u \le c/3$, the function $j(j+1)/(2(cj + u)^2)$ decreases for $j \ge 1$: its derivative has the sign of $j(2u - c) + u \le 0$. Its maximum is $1/(c + u)^2$. For $u > c/3$, lower u to $c/3$ in this bound. The positive and negative

crossing events are disjoint. Hence the two quadratic prices use a maximum, not their sum. When $R = 0$ or -1, both costs vanish.

For $h \geq 3$, put $j = h - 1 \geq 2$. Since $u \geq -1/2$,

$$\frac{\binom{h}{3}}{[c(h-1)+u]^3} \leq \frac{j^3 - j}{6(cj - 1/2)^3} \leq \frac{1}{(2c - 1/2)^3}.$$

For the last inequality the derivative numerator has the sign of $2cj - (3j^2 - 1)/2 < 0$, using $c \leq 1$ and $j \geq 2$; equality of the last bound occurs at $j = 2$. This proves the cubic estimate. On this same event $V_k > d$, so $(V_k)_+^3/d^3 \leq (V_k)_+^4/d^4$. For the other values of R, the left side is zero. Thus the quartic estimate is pointwise too; no approximation at a crossing atom is used.

Independence and centering give, with variances s_i of the n terms,

$$\mathbb{E}V_k^4 = \sum_i \mathbb{E}Z_i^4 + 6\sum_{i<j} s_i s_j \leq A^2 t + 3t^2 \leq t(A^2 + 3T).$$

Cauchy–Schwarz also gives $\mathbb{E}(V_k)_+^3 \leq t\sqrt{A^2 + 3T}$. Take expectations in the three pointwise estimates and then in (14). The result includes $t = 0$. □

The fourth-moment price is smaller whenever $A^2 + 3T < (2c - 1/2)^2$. It uses a moment bound already present in the old proof. In that domain the square root is unnecessary for this term. For larger noise retain the cubic price; taking the minimum preserves both regimes. Dropping the individual fourth moments $A^2 t$ would be false, even for a centered three-point variable with rare nonzero values.

6.1 The phase maximum still has three candidates

In a closed scalar cell, $\chi \leq b + u$. Define

$$g(u) = \max\left\{\frac{(b+u-1)_+}{(c-u)^2},\ \frac{(b+u)_+}{[c + \min(u, c/3)]^2}\right\}, \qquad u_* = c - 2b.$$

For $[l, h] \subseteq [-1/2, 1/2]$, its exact maximum is

$$\max\left\{g(l), g(h),\ \frac{1}{4(c-b)} \;\middle|\; c > b,\ l \leq u_* \leq h,\ u_* \leq c/3\right\}. \tag{18}$$

An ineligible third value is omitted. To prove this, the first summand inside the maximum is nondecreasing: on its positive region its derivative has positive numerator $c+u+2b-2 > c-u > 0$. Below $c/3$, the positive part of the second has derivative $(c - 2b - u)/(c + u)^3$; above $c/3$ it is nondecreasing. Its only possible interior maximum is therefore u_*, with value $1/[4(c-b)]$. At that point $b + u_* = (c + u_*)/2 \leq 3/4$, so the first term vanishes. The supremum of a maximum is the maximum of the two suprema. This proves (18), including singleton intervals, kinks and $c = 1$.

For every allowed parameter, g is bounded by the old sum of these two terms. Both terms are nonincreasing in c. Consequently a conservative cache at a smaller acquired c is still valid, and the new exact phase maximum is no larger than either the previous exact sum maximum or its 64-cell outward bound.

6.2 What this changes in the local comparison

Use the same acquired amplitude A, upper own-count variance T and lower gap coefficient c as the current cell. Replace its curvature coefficient by (17), maximizing the phase by (18). All other terms, including the marked-crossing linear term $\pi_k \mathbb{E}R$, the Gaussian repair and the actual binomial mass bound, are retained. The new coefficient is never larger than the old one. When

passing from $1/n$ to $1/N$, clamp the complete improved inverse-index correction at zero; its old strict positivity need not survive an improvement. The existing negative event bracket is still required before using a lower binomial mass bound.

For a fixed noise band, A, T, c are frozen valid bounds. Hence this replacement changes only the constant part of the inverse-index correction; the existing convex endpoint argument in the noise parameter is unchanged. It does not prove the acquisition of those bounds, cover all count cells, or replace the separate zero-noise binomial theorem. It supplies a smaller curvature contribution under exactly the old moment and coupling hypotheses.

7 One response for the actual moment polynomial

We give a conditional, uniform improvement on a recorded central probability panel. It bounds all the moment terms of the fixed-target raw remainder together. The original gap integrals and the Fourier remainder remain explicit: the statement does not establish the acquisition of this panel for a fixed-target maximizer, or a complete local exclusion.

Let P have mean zero, variance one and bounded support, and put $p = P(X > 0)$, $q = 1 - p$, $m = \mathbb{E}X_+ = \mathbb{E}X_-$. Assume a central support gap $(-A, B)$, with $A, B > 0$. Write $V = \mathbb{E}[X \mid \mathbf{1}_{X>0}]$, $W = X - V$, $a = m/q$, $b = m/p$, and $\theta = \mathbb{E}W^2$. Thus $V = -a$ or b, and $m^2 = (1-\theta)pq$. Use the same actual gap quantities J, J_1, J_2 as in the conserved identity, and write $\sigma = \sqrt{10}$, $p_* = (4-\sigma)/2$ and $D = (3+\sigma)\rho - \gamma - 3J$. For a law Y in the original physical units, denote its moments by

$$(\rho_Y, \beta_Y, \gamma_Y, M_Y, E_Y, Z_Y) = (\mathbb{E}|Y|^3, \mathbb{E}Y|Y|, \mathbb{E}Y^3, \mathbb{E}Y^4, \mathbb{E}|Y|^5, \mathbb{E}Y|Y|^3).$$

The unsubscripted moments refer to X, and subscript 0 refers to V. In particular, V has variance $1 - \theta$, not necessarily one.

Fix $c = C_{\mathrm{E}} = (3+\sqrt{10})/(6\sqrt{2\pi})$ and $\varphi = \phi(H/\sqrt{n})$, where ϕ is the standard normal density and H is the raw threshold. For any moment tuple M define the polynomial

$$\begin{aligned} T_c(M; J, J_1, J_2, H, \varphi) = c\left\{\frac{3}{2}\beta J + 3\rho J_1 - 3J_2 - \frac{E}{4} + \frac{\rho M_4}{2} + \frac{3}{2}\beta\gamma - \frac{\rho}{2}\right\} \\ + H\left\{c\left(Z - 3\beta - \frac{3}{2}\rho\gamma\right) + \varphi\left(J_1 + \frac{M_4 - 3}{6}\right)\right\} - H^2(c\rho + \varphi\gamma/2). \end{aligned}$$

Thus the actual raw remainder is $\mathcal{R}_c(H) = T_c(M(P); J(P), J_1(P), J_2(P), H, \varphi) + E_n(H)$. Every comparison below keeps the actual $J(P), J_1(P), J_2(P)$ on both sides. Evaluating this polynomial at $M(V)$ is not evaluating the remainder of a Bernoulli law.

7.1 Exact common-measure identity

On the two sign classes put

$$d\mu_+ = W^2\mathbf{1}_{X>0}\,dP, \qquad d\mu_- = W^2\mathbf{1}_{X<0}\,dP.$$

For magnitude $x > 0$ and conditional mean magnitude $v > 0$, define

$$Q_3 = x + 2v, \quad Q_4 = x^2 + 2vx + 3v^2, \quad Q_5 = x^3 + 2vx^2 + 3v^2x + 4v^3.$$

The identity $x^j - v^j - jv^{j-1}(x - v) = (x - v)^2 Q_j(v, x)$, together with conditional centering, gives the moment differences. For example, $\delta\rho = \sum_{\pm}\int Q_3\,d\mu_{\pm}$, $\delta\gamma = \int Q_3\,d\mu_+ - \int Q_3\,d\mu_-$, and $\delta\beta = \mu_+(\mathbb{R}) - \mu_-(\mathbb{R})$.

For $\epsilon \in \{+1, -1\}$, with $v = b$ or a respectively, put

$$\begin{aligned}
P_{0,\epsilon}(v, x) &= -\frac{Q_5}{4} + \frac{\rho}{2} Q_4 + \left(\frac{M_0}{2} - \frac{1}{2} + 3J_1 + \frac{3}{2}\epsilon\beta \right) Q_3 + \frac{3}{2}\epsilon(\gamma_0 + J), \\
P_{1,\epsilon}(v, x) &= \epsilon(Q_4 - 3) - \frac{3}{2}(\epsilon\rho + \gamma_0) Q_3, \\
\mathscr{P}_\epsilon(v, x) &= cP_{0,\epsilon} + H \left(cP_{1,\epsilon} + \frac{\varphi}{6} Q_4 \right) - H^2(c + \epsilon\varphi/2) Q_3.
\end{aligned}$$

Then the following identity is exact:

$$T_c(M(P); J, J_1, J_2, H, \varphi) - T_c(M(V); J, J_1, J_2, H, \varphi) = \sum_{\epsilon=\pm} \int \mathscr{P}_\epsilon \, d\mu_\epsilon. \tag{19}$$

To check the products, use

$$\rho M_4 - \rho_0 M_0 = \rho\delta M_4 + M_0 \delta\rho, \quad \beta\gamma - \beta_0\gamma_0 = \beta\delta\gamma + \gamma_0\delta\beta, \quad \rho\gamma - \rho_0\gamma_0 = \rho\delta\gamma + \gamma_0\delta\rho.$$

The coefficients contain actual moments of the same law. No independent choice of incompatible moment maxima is asserted to be a law.

7.2 The cost of the common measures

Let $\Delta_A = a - A$, $\Delta_B = b - B$ and define

$$B_0 = \frac{(6 + 2\sigma)(1 - \theta)^{3/2}}{\sqrt{pq}} (p - p_*)^2, \quad E_0 = 6m\{(A + B)(\Delta_A + \Delta_B) + \Delta_A \Delta_B\}.$$

The weighted gap inequality gives, for $r = D - B_0 - E_0$,

$$0 \le \sum_{\pm} \int k_\pm \, d\mu_\pm \le r, \quad k_+(x) = (\sigma - 1)(x + 2b) + 6B, \quad k_-(x) = (\sigma + 1)(x + 2a) + 6A. \tag{20}$$

For completeness, the outer-gap estimate implies

$$D \ge (\sigma - 1)\mathbb{E}X_+^3 + (\sigma + 1)\mathbb{E}X_-^3 - 6mAB + 6BR_+ + 6AR_-, \quad R_+ = \mathbb{E}X_+^2 - mB, \; R_- = \mathbb{E}X_-^2 - mA.$$

Substitute $\mathbb{E}X_+^3 = mb^2 + \int (x + 2b) \, d\mu_+$, $R_+ = \mu_+(\mathbb{R}) + m\Delta_B$, and the reflected identities. The deterministic part is $B_0 + E_0$, by $(\sigma + 1)p^2 + (\sigma - 1)q^2 - 6pq = (6 + 2\sigma)(p - p_*)^2$. This proves (20). All costs and both measures are nonnegative. In particular a pointwise bound $\mathscr{P}_\pm \le K k_\pm$, with $K \ge 0$, spends this residual only once.

8 A uniform bound on the recorded central panel

The domain is taken from zero-based central row 19 of the retained confinement table. Its acquisition is a premise; historical success metadata is not a fresh verification. Assume simultaneously

$$\begin{aligned}
&\frac{209}{500} \le p \le \frac{21}{50}, \quad |X| \le \frac{46}{25}, \quad \theta \le \frac{12657}{10^6}, \quad \rho \le \frac{214889}{200000}, \\
&\frac{72659}{500000} \le \beta \le \frac{174581}{10^6}, \qquad A \ge \frac{810731}{10^6}, \quad B \ge \frac{284079}{250000}.
\end{aligned} \tag{21}$$

These are closed-domain bounds for one actual law.

Theorem 8.1. *Under* (21)*, for every integer* $n \geq 1$ *and every real* H*,*

$$\mathcal{R}_c(H) \leq T_c(M(V); J(P), J_1(P), J_2(P), H, \varphi) + c\left(\frac{221}{200} + \frac{9}{50}|H|\right) r + E_n(H). \tag{22}$$

Here $c = C_{\mathrm{E}}$*,* $\varphi = \phi(H/\sqrt{n})$*, and* r *is the exact same-law residual in* (20)*. The conclusion bounds the moment polynomial; it does not supply a Fourier bound for* E_n *or replace the actual gaps by those of* V*.*

Proof. We first give rational consequences of the acquired domain:

$$\begin{gathered} a \in [21/25, 213/250], \quad b \in [29/25, 119/100], \quad m > 49/100, \\ A > 81/100, \ B > 113/100, \quad 1 \leq \rho < 43/40, \quad 29/200 < \beta < 7/40, \\ 3/10 < \gamma_0 < 1/3, \quad M_0 < 10/9, \quad J < 83/40, \quad \tfrac{3}{2}(\gamma_0 + J) > 3, \quad J_1 < 1. \end{gathered}$$

The conditional mean bounds follow by squaring $a^2 = (1-\theta)p/q$ and $b^2 = (1-\theta)q/p$. For $v = pq$, its interval is $[60819/250000, 609/2500]$. The identities

$$\gamma_0 = (1-\theta)^{3/2}(1-2p)/\sqrt{v}, \qquad M_0 = (1-\theta)^2(1/v - 3)$$

give the two skeleton bounds; squaring positive quantities suffices. Moreover $J \leq \rho + 2m \leq \rho + 1$, whereas $J \geq m(A+B)^2$. For the last bound, negative-threshold portions of J_1 are nonpositive. On positive outer gaps, $t - u \leq t$; on the central positive portion $t - u = t + A$. Consequently

$$\begin{aligned} J_1 &\leq \int_0^\infty t^2 \tau(t)\, dt + \frac{AmB^2}{2} = \frac{\mathbb{E}X_+^4}{3} + \frac{AmB^2}{2} \\ &< \frac{(46/25)(43/40)}{3} + \frac{(213/250)(119/100)^2}{4} = \frac{28828879}{30000000} < 1. \end{aligned}$$

The positive integrands justify summing countably many gaps. This uses the acquired support before the fourth moment is used.

Since $\sigma > 79/25$, define lower costs

$$d_+(x) = \frac{14739}{1250} + \frac{54}{25}x, \qquad d_-(x) = \frac{14811}{1250} + \frac{104}{25}x.$$

They are strictly positive and at most $k_+(x), k_-(x)$ respectively. The constant responses are bounded by

$$\begin{aligned} P_{0,+} \leq p_+(x) &= \frac{435879881}{36000000} + \frac{320027}{90000}x - \frac{17}{400}x^2 - \frac{1}{4}x^3, \\ P_{0,-} \leq p_-(x) &= \frac{895536247}{375000000} + \frac{289067}{90000}x + \frac{47}{400}x^2 - \frac{1}{4}x^3. \end{aligned}$$

Indeed use $\rho/2 \leq 43/80$ and $M_0/2 - 1/2 + 3J_1 \leq 55/18$. On the positive class use $3\beta/2 \leq 21/80$ and $3(\gamma_0 + J)/2 \leq 289/80$; on the negative class use $-3\beta/2 \leq -87/400$ and $-3(\gamma_0 + J)/2 \leq -3$. For the resulting cubic, the x^2 and x coefficients decrease with the mean magnitude, and the constant coefficient increases. Inserting the corresponding endpoints gives exactly these two polynomials.

There is a finite, exact check of their ratio maxima. For a cubic p and affine $d = d_0 + d_1 x$, its derivative numerator is

$$G(x) = p_1 d_0 - p_0 d_1 + 2p_2 d_0 x + (p_2 d_1 + 3p_3 d_0)x^2 + 2p_3 d_1 x^3.$$

For each class, $G'' < 0$ and G' is negative at the left endpoint; thus G strictly decreases. Its signs at the two endpoints are opposite, so precisely one stationary maximum must be checked. Rational enclosing brackets and strict bounds are

	x domain	x at the stationary maximum	$p(x)/d(x)$ upper bound
$+$	$[113/100, 46/25]$	$[1194059, 1194060]/10^6$	$221/200$
$-$	$[81/100, 46/25]$	$[1662099, 1662100]/10^6$	$3/8$.

The endpoint values are smaller as well. These are rational inequalities: on a positive interval $[l, u]$, bound each monomial $p_j x^j$ using u when $p_j \geq 0$ and l otherwise, then divide by $d(l) > 0$. The accompanying checker prints all exact coefficients, derivative signs, brackets and endpoint comparisons; no sampled positivity is used.

For the linear term in H, use $0 \leq \varphi/c \leq 6/(3+\sqrt{10}) < 1$. After division by $c|H|$, the following four polynomials majorize its response, on the same respective magnitude domains:

$\operatorname{sgn} H$	$\operatorname{sgn} X$	upper polynomial $s(x)$
$+$	$+$	$-53693/20000 + (62/75)x + (7/6)x^2$
$+$	$-$	$3189/1000 - (19/80)x - (5/6)x^2$
$-$	$+$	$19321/5000 - (83/400)x - x^2$
$-$	$-$	$-157893/62500 + (88/125)x + x^2$.

For example, the first is obtained from $(7/6)Q_4 - 3 - (39/20)Q_3$, maximizing over $b \leq 119/100$. The others follow from $-(5/6)Q_4 + 3 + (93/80)Q_3$, $-Q_4 + 3 + (169/80)Q_3$, and $Q_4 - 3 - Q_3$, respectively. The mean-endpoint directions follow by differentiating these displayed quadratics with respect to their mean magnitude. For each row, $(9/50)d - s > 0$ throughout the closed interval: if its quadratic coefficient is negative, check both endpoints; otherwise its derivative is positive from the left endpoint, where its value is positive. All these endpoint tests are exact rational inequalities.

Finally $\varphi < c$ implies $c > \varphi/2$, so the H^2 term of each $\mathscr{P}_\epsilon$ is nonpositive. We have proved pointwise

$$\mathscr{P}_\pm(v, x) \leq c\,(221/200 + (9/50)|H|)\,k_\pm(x).$$

Integrate once against the two positive measures and use (19)–(20). □

9 What improves, and what remains

The gain is more than a new notation for the positive-measure lemma. At $H = 0$, consider the explicit split-response bound that discards $-Q_5/4 \leq 0$ before taking the positive-class supremum, and uses the same upper bounds for all remaining coefficients. Its required constant is at least its value at $x = 46/25$, $b = 119/100$:

$$K_{\text{split}} \geq \frac{(43/80)Q_4 + (55/18 + 21/80)Q_3 + 289/80}{d_+(46/25)} = \frac{173316497}{113512320}.$$

In contrast, the combined constant is $221/200$, smaller by at least $239426917/567561600 > 0.4218$. For $r > 0$ this gives a strict improvement of that uniform upper bound. This benchmark is an explicitly defined relaxation, not a claim that any law realizes its independent moment endpoints, or that this beats every previously implemented atlas bound.

The domain has actual laws with positive noise. For example the masses

$$(29/100, 29/100, 21/100, 21/100) \quad \text{at} \quad (-533, -517, 717, 733)/617$$

give mean zero, variance one, and conditional means $-525/617, 725/617$. Their moments and support satisfy (21), with $0 < \theta < 12657/10^6$; exact checks are recorded. This law is only a non-vacuity and identity control, not a claimed maximizing witness.

To turn (22) into a complete raw bound, one must still control the baseline polynomial with the original $J(P), J_1(P), J_2(P)$ and the full signed Fourier term $E_n(H)$. Conditional mean replacement changes the gaps, and V is not standardized when $\theta > 0$. Consequently neither the Bernoulli theorem nor a putative identity $J(P) = J(V)$ supplies these missing steps. The independent conditional expectation response also does not remove the quadratic remainder in $f^n - g^n - ng^{n-1}(f - g)$ or provide high-frequency decay. Finally, the old table concerns its original attained-ratio acquisition; using its domain for a fixed-target maximizer requires the separately stated same-law acquisition/transfer hypotheses. No certificate or cell is removed by this result.

Research and verification record. The accompanying source files contain exact checkers, source locations, input hashes, failed controls and fresh execution records. Independent review rederived the count-cost conclusion without importing its constructor. The coordinating review checked the entry and response derivations and the order of the numerical implications. These are scoped mathematical reviews, not a formalization or an end-to-end certificate acceptance. Existing certificates and their original targets are preserved.

Use of AI. AI systems were used extensively in developing and checking the arguments, writing the programs, and drafting this note. The proofs and limitations are presented for author and independent mathematical review; agreement between AI systems is not evidence beyond the arguments and checks shown.